SESSION 8 · THURSDAY JULY 9, 2026 · PHASE 2 — THE BRIDGE · IN-CLASS

✓ ANSWER SHEET · TEACHER / SELF-CHECK

Activity Sheet — Answer Key

Same Logic, Two Languages: Scratch → Python · Follow the Daily Rhythm: warm-up → mini-lesson → guided build → open build → share-out

WARM-UP 10 MIN · MINI-LESSON 20 MIN · GUIDED BUILD 20 MIN · OPEN BUILD 30–40 MIN · SHARE-OUT 10 MIN

YOUR NAME	TODAY'S DATE	SECTION / PERIOD
First Last	Jul 9, 2026	

WARM-UP PART 1 · PUZZLE 1 — SCRATCH → PYTHON (ANSWERS)

Fill the four Python blanks to match the Scratch script:

```python
# runs top to bottom!
print("Hello, World!")
for i in range(3):
    print("I love coding!")
print("Done!")
```

Blank 1 → `"Hello, World!"` — matches Scratch's first say block
Blank 2 → `3` — Scratch's repeat 3 = `range(3)`
Blank 3 → `"I love coding!"` — loop body, **indented 4 spaces**
Blank 4 → `"Done!"` — NOT indented; runs once after loop

☆ The last print() is at the **same indent level** as the `for` keyword — it's outside the loop, just like code after a Scratch repeat block!

WARM-UP PART 1 · PUZZLE 2 — PREDICT THE OUTPUT (ANSWERS)

```python
print("Countdown:")
for i in range(3):
    print(i)
print("Blast off!")
```

Line	Output	Why
1	`Countdown:`	plain print(), runs once before loop
2	`0`	i = 0 on 1st iteration
3	`1`	i = 1 on 2nd iteration
4	`2`	i = 2 on 3rd (last) — range(3) stops before 3
5	`Blast off!`	NOT indented → runs after loop

i on 1st loop: `0` · **i on last loop:** `2`
⚠ `range(3)` gives **0, 1, 2** — always starts at 0, stops BEFORE stop value!

MINI-LESSON PART 2 · VOCABULARY — ANSWERS

WORD	WHAT IT MEANS	SCRATCH CONNECTION
`print()`	Displays text or a value in the output console	say [ ] for (2) secs — both show a message
`string`	A sequence of characters wrapped in "double quotes"	Text inside a say or join block
`for loop`	Repeats a code block a set number of times using a counter	repeat [ ] block — same concept, text syntax
`indentation`	4 spaces that tell Python a line is inside the loop	Blocks snapping inside the C-shaped repeat block
`syntax`	Grammar rules of a language — must be exact or error	Scratch blocks only snap together correctly; Python enforces this via SyntaxError

Anatomy — labeled:

```python
for i in range(5):    print("Hello!")
```

AI MOMENT — DISCUSSION ANSWER

```python
print("I like ... ")
print("learning ... ")
print("new things")
# AI sees the pattern → predicts next
```

? If an AI learned only from your writing, what patterns would it predict? Would it be a good AI for everyone?

Model full-credit answer (1–2 sentences):
"If an AI learned only from my writing, it would predict the words, topics, and sentence structures I use most. It would be a good AI for me, but not for everyone — it would be biased toward my age, language, and experiences."

Key ideas to award credit for:
1. AI finds patterns in training data
2. Limited data → limited/biased AI
3. "Good for everyone" needs diverse, large-scale data

Part	Name	Role
`for`	keyword	Starts the loop
`i`	loop variable	Holds current count (0,1,2,3,4)
`range(5)`	range function	Generates sequence 0–4
`:`	colon	Opens the loop body — required!
`    print()`	indented body	Runs on every iteration

🏗 GUIDED BUILD PART 3 · EVERYONE TOGETHER — ANSWERS

Complete expected program (after all 5 steps):

```
print("My name is Alex")
print("I am 10 years old")
print("I love coding")
# now a loop
for i in range(5):
    print(i)
```

#	Verify	Answer
1	3 prints run?	✅ Quotes · lowercase print · closing paren
2	Why use comments?	Comments explain code to humans — Python ignores lines starting with `#`
3	What does the colon mean?	Signals start of loop body — Python raises SyntaxError without it
4	4-space indent on print(i)	Exactly 4 spaces — wrong indent → IndentationError
5	i on 3rd loop?	Output: 0,1,2,3,4 · **i on 3rd loop = 2** (0-indexed)
6	🐛 Bug Hunt	Missing colon → SyntaxError · No indent → IndentationError · Missing quotes → NameError

🎖 OPEN BUILD PART 4 · TIER TASK — ANSWERS

🌍 TIER A · Gr 1–3

```
print("Lions are big cats")
print("They live in Africa")
print("They roar loudly")
for i in range(3):
  print("Roar!")
```

Loop output:
"Roar!" × 3
i = 0, 1, 2

💎 TIER B · Gr 3–5

```
# About Me
print("My name is Sam")
print("I am 9 years old")
print("I like soccer")
print("My goal: code daily")
# Counter loop
for i in range(1,6):
  print(i)
```

range(5) →
0,1,2,3,4
range(1,6) →
1,2,3,4,5
i on loop 3 = **3**

💜 TIER C · Gr 6–9

```
for i in range(0,20,2):
  print(i, "is even")
```

range(0,20,2) →
0,2,4,...,18
range(0,10,3) →
0,3,6,9
Python: variables, files, math — Scratch can't easily do these

⚡ CHALLENGE CARD — COUNTDOWN ANSWERS

```
# Countdown 10 → 1
for i in range(10, 0, -1):
    print(i)
print("Blast off! 🚀")  # NOT indented!
```

What does -1 (step) tell Python?
Count **down** by 1 each iteration: 10→9→8→...→1. A negative step = countdown.

Why is "Blast off!" NOT indented?
It's outside the loop block — it runs exactly **once**, after all 10 iterations finish. Same as code placed after a Scratch repeat block.

✨ SHARE-OUT PART 5 · PAIR SHARE + DEBUG CORNER

"My loop runs **_N_ times** and prints **_value_** each time."
Partner: i the very first time? → **i = 0** (range always starts at 0 unless given a start arg)

Bug	Error Type	Fix
`print(Hello)`	NameError	`print("Hello")`
`for i in range(5)`	SyntaxError	Add `:` at end
Body not indented	IndentationError	Add 4 spaces before body line

✍ EXIT TICKET — QUICK CHECK — MODEL ANSWERS

1. What does `print()` do? (one sentence)

`print()` displays text or a variable's value in the output console so the user can see it.

2. What does the indent (4 spaces) tell Python?

The indent tells Python that this line is **inside the loop** and should run on every iteration. Without it, Python raises an `IndentationError`.

3. One thing Python and Scratch have in common:

Both run code **top to bottom (sequentially)**, both support loops/ repetition, both can display messages. (Any one = full credit)

😊 How do you feel? → Give credit for any honest circle
🚀 "One thing I still want to learn" → any answer = full credit

📊 SCORING RUBRIC

SECTION	WHAT TO LOOK FOR	PTS
🧩 Warm-Up Puzzles	Both puzzles; output predicted; blanks filled	3
⌨️ Vocab + Annotate	All 5 vocab rows; anatomy labels done	3
🤖 AI Moment	Pattern + bias/diversity idea present	1
🏗 Guided Build	Code complete; predictions made; error box filled	4
🎖 Tier Task	Tier task complete + running; checklist ✅	5
🐛 Debug Corner	Bug named + fix described	1
✍ Exit Ticket	All 3 questions answered; feeling circled	3
⚡ Challenge Card	Countdown built; both questions answered	+3
Required total / With challenge bonus		20 / 23

📅 **Homework Due Mon Jul 13:** Type a 4-line "About Me" Python program using `print()`. Run it. Fix any errors. 3 tiers available.
Next Session Mon Jul 13: Python Variables, `input()` & a Mad Libs Generator! 🎉

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